

Data Loading (ETL)

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Assignment Name: Data Loading (ETL)

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Dataset:

```
CREATE DATABASE etl_loading_db;  
USE etl_loading_db;
```

```
CREATE TABLE Sales_Data (  
    Order_ID VARCHAR(10),  
    Customer_ID VARCHAR(10),  
    Sales_Amount VARCHAR(20),  
    Order_Date VARCHAR(20)  
);
```

```
INSERT INTO Sales_Data  
(Order_ID, Customer_ID, Sales_Amount, Order_Date)  
VALUES  
( 'O101','C001','4500','12-01-2024'),  
( 'O102','C002',NULL,'15-01-2024'),  
( 'O103','C003','3200','2024/01/18'),  
( 'O101','C001','4500','12-01-2024'),  
( 'O104','C004','Three Thousand','20-01-2024'),  
( 'O105','C005','5100','25-01-2024');
```

Q1: Identify all data quality issues present in the dataset that can cause problems during data loading.

Answer :

```
SELECT * FROM Sales_Data;
```

The dataset contains the following data quality issues:

1. Duplicate record
 - Order_ID = O101 appears twice.
2. Missing value
 - Sales_Amount is NULL for Order_ID O102.
3. Invalid numeric value
 - "Three Thousand" is stored as text instead of a numeric value.
4. Inconsistent date format
 - Most dates use DD-MM-YYYY.
 - One record uses YYYY/MM/DD.
5. Sales_Amount stored as VARCHAR instead of a numeric datatype.

```
22 • SELECT * FROM Sales_Data;
23
```

	Order_ID	Customer_ID	Sales_Amount	Order_Date
▶	O101	C001	4500	12-01-2024
	O102	C002	NULL	15-01-2024
	O103	C003	3200	2024/01/18
	O101	C001	4500	12-01-2024
	O104	C004	Three Thousand	20-01-2024
	O105	C005	5100	25-01-2024

Q2: Assume Order_ID is the Primary Key.

a) Is the dataset violating the Primary Key rule?

b) Which record(s) cause this violation?

Answer :

```
SELECT
```

```
    Order_ID,
```

```
    COUNT(*) AS Occurrences
```

```
FROM Sales_Data
```

```
GROUP BY Order_ID
```

```
HAVING COUNT(*) > 1;
```

a) Yes. The dataset violates the Primary Key rule.

b) Order_ID O101 appears twice.

```
24 • SELECT
25     Order_ID,
26     COUNT(*) AS Occurrences
27 FROM Sales_Data
28 GROUP BY Order_ID
29 HAVING COUNT(*) > 1;
30
```

	Order_ID	Occurrences
▶	O101	2

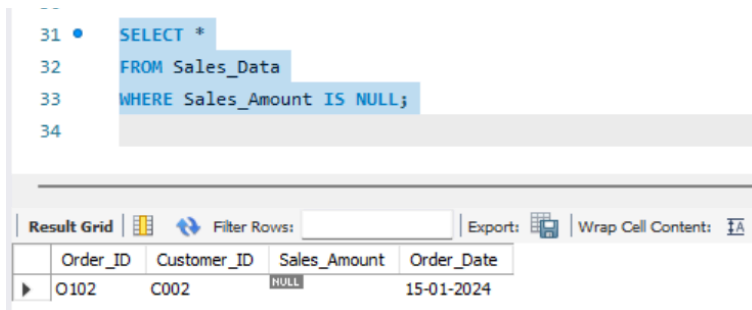
Q3. Missing Value Analysis

Which column(s) contain missing values?

- a) List the affected records.
- b) Explain why loading these records without handling missing values is risky.

Answer :

```
SELECT *  
FROM Sales_Data  
WHERE Sales_Amount IS NULL;
```



a) Affected Records

The Sales_Amount column contains missing values.

Affected record:

Order_ID	Customer_ID	Sales_Amount	Order_Date
O102	C002	NULL	15-01-2024

b) Why is loading these records without handling missing values risky?

Loading records with missing values is risky because:

1. Sales reports and KPIs may become inaccurate.
2. Aggregate calculations such as SUM(), AVG(), and revenue analysis may produce incorrect or incomplete results.
3. If Sales_Amount is a mandatory field, the data load may fail due to constraint violations.
4. Missing values can lead to poor business decisions based on incomplete data.

Q4: Identify records where Sales_Amount violates expected data type rules.

a) Which record(s) will fail numeric validation?

b) What would happen if this dataset is loaded into a SQL table with Sales_Amount as DECIMAL?

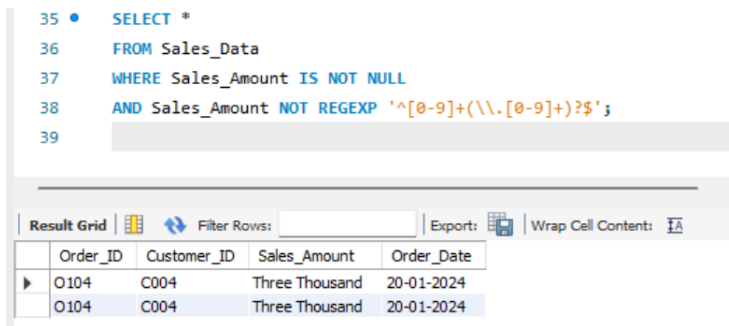
Answer :

SELECT *

FROM Sales_Data

WHERE Sales_Amount IS NOT NULL

AND Sales_Amount NOT REGEXP '^[0-9]+(\\.[0-9]+)?\$';



Record failing validation:

- Order_ID = O104
- Sales_Amount = "Three Thousand"

If loaded into a DECIMAL column:

- The load may fail, or
- The invalid value may become NULL (depending on SQL mode), causing inaccurate data.

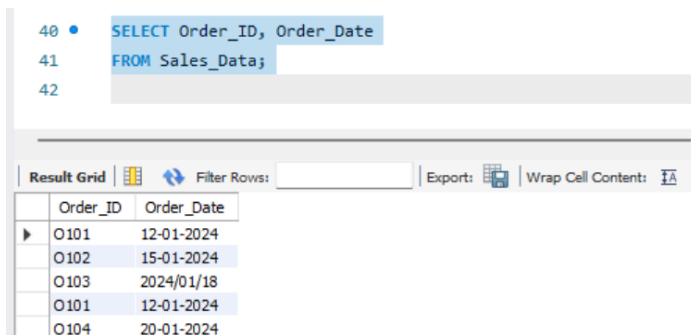
Q5: The Order_Date column has multiple formats.

- a) List all date formats present.
- b) Why is this a problem during data loading?

Answer :

```
SELECT Order_ID, Order_Date
```

```
FROM Sales_Data;
```



Order_ID	Order_Date
O101	12-01-2024
O102	15-01-2024
O103	2024/01/18
O101	12-01-2024
O104	20-01-2024

Formats found:

- DD-MM-YYYY
- YYYY/MM/DD

Problem:

Mixed formats can cause parsing errors, incorrect dates, or ETL failures because the loader expects a consistent format.

Q6. Load Readiness Decision

Based on the dataset condition:

- a) Should this dataset be loaded directly into the database?
- b) Justify your answer with at least three reasons.

Answer :

```
SELECT * FROM Sales_Data;
```

43 • `SELECT * FROM Sales_Data;`
44

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Order_ID	Customer_ID	Sales_Amount	Order_Date
O101	C001	4500	12-01-2024
O102	C002	NULL	15-01-2024
O103	C003	3200	2024/01/18
O101	C001	4500	12-01-2024
O104	C004	Three Thousand	20-01-2024
O105	C005	5100	25-01-2024
O101	C001	4500	12-01-2024
O102	C002	NULL	15-01-2024
O103	C003	3200	2024/01/18
O101	C001	4500	12-01-2024
O104	C004	Three Thousand	20-01-2024
O105	C005	5100	25-01-2024

a) No, this dataset should **not** be loaded directly into the database.

b) **Reasons:**

1. The dataset contains duplicate Order_ID values (O101).
2. The Sales_Amount column contains a missing (NULL) value.
3. The Sales_Amount column contains an invalid text value ("Three Thousand") instead of a numeric value.
4. The Order_Date column contains inconsistent date formats.

These issues should be cleaned before loading to ensure data quality.

Q7. Pre-Load Validation Checklist

List the exact pre-load validation checks you would perform on this dataset before loading.

Answer :

The following validations should be performed before loading:

1. Check for duplicate primary keys.

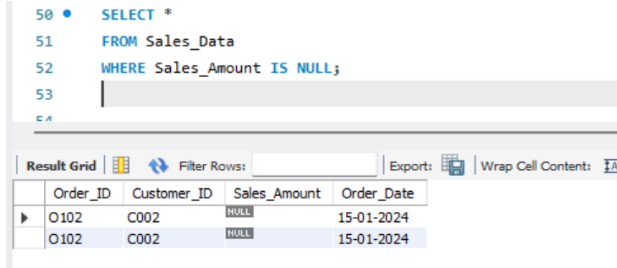
```
SELECT Order_ID, COUNT(*) AS Count
FROM Sales_Data
GROUP BY Order_ID
HAVING COUNT(*) > 1;
```

45 • `SELECT Order_ID, COUNT(*) AS Count`
46 `FROM Sales_Data`
47 `GROUP BY Order_ID`
48 `HAVING COUNT(*) > 1;`
49
50

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Order_ID	Count		
O101	4		
O102	2		
O103	2		
O104	2		
O105	2		

2. Check for missing values.

```
SELECT *  
FROM Sales_Data  
WHERE Sales_Amount IS NULL;
```

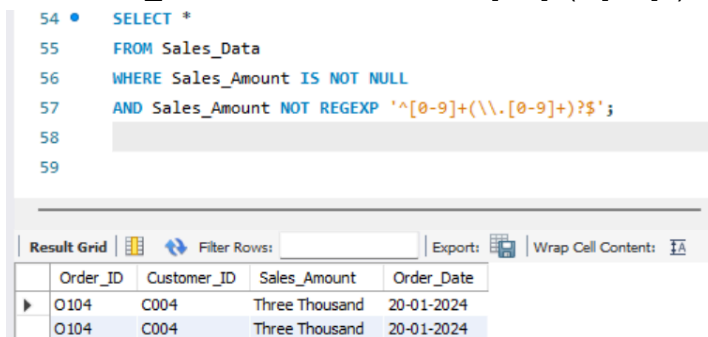


```
50 • SELECT *  
51 FROM Sales_Data  
52 WHERE Sales_Amount IS NULL;  
53  
54
```

Order_ID	Customer_ID	Sales_Amount	Order_Date
O102	C002	NULL	15-01-2024
O102	C002	NULL	15-01-2024

3. Validate numeric fields.

```
SELECT *  
FROM Sales_Data  
WHERE Sales_Amount IS NOT NULL  
AND Sales_Amount NOT REGEXP '^[0-9]+(\\.[0-9]+)?$';
```

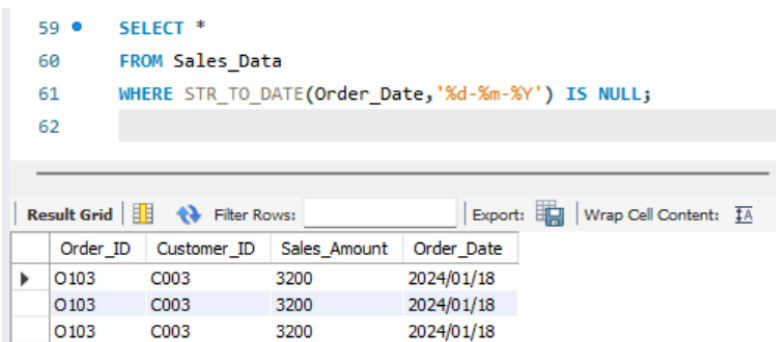


```
54 • SELECT *  
55 FROM Sales_Data  
56 WHERE Sales_Amount IS NOT NULL  
57 AND Sales_Amount NOT REGEXP '^[0-9]+(\\.[0-9]+)?$';  
58  
59
```

Order_ID	Customer_ID	Sales_Amount	Order_Date
O104	C004	Three Thousand	20-01-2024
O104	C004	Three Thousand	20-01-2024

4. Validate date formats.

```
SELECT *  
FROM Sales_Data  
WHERE STR_TO_DATE(Order_Date,'%d-%m-%Y') IS NULL;
```



```
59 • SELECT *  
60 FROM Sales_Data  
61 WHERE STR_TO_DATE(Order_Date,'%d-%m-%Y') IS NULL;  
62
```

Order_ID	Customer_ID	Sales_Amount	Order_Date
O103	C003	3200	2024/01/18
O103	C003	3200	2024/01/18
O103	C003	3200	2024/01/18

5. Blank Value Validation

```
SELECT *  
  
FROM Sales_Data  
  
WHERE TRIM(Order_ID) = "  
  
OR TRIM(Customer_ID) = "  
  
OR TRIM(Sales_Amount) = "  
  
OR TRIM(Order_Date) = ";
```

```
63 • SELECT *  
64 FROM Sales_Data  
65 WHERE TRIM(Order_ID) = ''  
66 OR TRIM(Customer_ID) = ''  
67 OR TRIM(Sales_Amount) = ''  
68 OR TRIM(Order_Date) = '';  
69
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Order_ID	Customer_ID	Sales_Amount	Order_Date
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6. Check business rules (e.g., Sales Amount > 0).

```
SELECT *  
  
FROM Sales_Data  
  
WHERE Sales_Amount REGEXP '^[0-9]+(\\.[0-9]+)?$'  
  
AND CAST(Sales_Amount AS DECIMAL(10,2)) <= 0;
```

```
70 • SELECT *  
71 FROM Sales_Data  
72 WHERE Sales_Amount REGEXP '^[0-9]+(\\.[0-9]+)?$'  
73 AND CAST(Sales_Amount AS DECIMAL(10,2)) <= 0;  
74  
75
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Order_ID	Customer_ID	Sales_Amount	Order_Date
----------	-------------	--------------	------------

Q8. Cleaning Strategy

Describe the step-by-step cleaning actions required to make this dataset load-ready.

Answer :

Step 1. Remove duplicate records (Order_ID O101).

Step 2. Handle the missing Sales_Amount value by imputing or requesting the correct value.

Step 3. Convert "Three Thousand" into the numeric value **3000**.

Step 4. Standardize all Order_Date values to a single format (YYYY-MM-DD).

Step 5. Change the Sales_Amount column datatype from VARCHAR to DECIMAL.

Step 6. Validate all cleaned records.

Step 7. Load the cleaned dataset into the database.

Q9. Loading Strategy Selection

Assume this dataset represents daily sales data.

a) Should a Full Load or Incremental Load be used?

b) Justify your choice.

Answer :

a) Incremental Load.

b) Reasons:

1. Daily sales data is generated continuously.
2. Reloading the entire dataset every day is inefficient.
3. Incremental loading processes only new or changed records.
4. It reduces execution time and resource usage.
5. It minimizes database load and improves ETL performance.

Q10. BI Impact Scenario

Assume this dataset was loaded without cleaning and connected to a BI dashboard.

- a)** What incorrect results might appear in the Total Sales KPI?
- b)** Which records specifically would cause misleading insights?
- c)** Why would BI tools not detect these issues automatically?

Answer :

a) Incorrect Results:

- Total Sales may be incorrect because:
 - Duplicate records increase the total.
 - NULL values reduce the calculated total.
 - Invalid text values cannot be summed correctly.

b) Misleading Records:

- **O101** – duplicated record.
- **O102** – missing Sales_Amount.
- **O104** – invalid Sales_Amount ("Three Thousand").

c) Reason:

BI tools generally visualize the data provided to them. They do not automatically identify duplicates, missing values, invalid data types, or inconsistent formats unless explicit validation and cleaning are performed during the ETL process.